The Nervous System: Neurons and Synapses

Human Physiology - Chapter 7

Biology Teaching

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Outline

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Introduction to Neurons

- **Neurons** (神经元) are the functional units of the nervous system
- Specialized for generating and conducting electrical signals
- Three main structural components:
  - **Cell body** (soma, 胞体): Metabolic center containing nucleus
  - **Dendrites** (树突): Receive input from other neurons
  - **Axon** (轴突): Conducts nerve impulses away from cell body

Neuron Structure: Motor and Sensory Types

- Motor neurons have dendrites extending from cell body
- Sensory neurons have pseudounipolar structure with single branching process
- Both types conduct electrical signals but differ in organization

Parts of a Neuron

- **Axon hillock** (轴丘): Where axon originates; site of action potential initiation
- **Myelin sheath** (髓鞘): Insulating layer formed by Schwann cells
- **Axon terminals** (轴突末梢): Branched endings forming synapses
- **Nodes of Ranvier** (郎飞结): Gaps in myelin where action potentials regenerate

CNS and PNS Organization

- **Central nervous system (CNS)** (中枢神经系统): Brain and spinal cord
- **Peripheral nervous system (PNS)** (周围神经系统): Nerves outside CNS
- **Sensory (afferent) neurons** (感觉神经元): Carry information to CNS
- **Motor (efferent) neurons** (运动神经元): Carry information from CNS to effectors

Neuron Classification by Structure

- **Pseudounipolar** (假单极): One process splits into two; typical sensory neurons
- **Bipolar** (双极): Two processes; found in retina and cochlea
- **Multipolar** (多极): Multiple dendrites and one axon; most common type
- Structure reflects functional specialization

Terminology

- **Nerves** (神经): Bundles of axons in PNS
- **Ganglia** (神经节): Clusters of cell bodies outside CNS
- **Nuclei** (神经核): Clusters of cell bodies within CNS
- **Tracts** (神经束): Bundles of axons within CNS
- Understanding terminology essential for neuroanatomy

Neuroglial Cells

- **Neuroglial cells** (glial cells, 神经胶质细胞) outnumber neurons
- Provide structural, metabolic, and protective support
- CNS glia: Oligodendrocytes, microglia, astrocytes, ependymal cells
- PNS glia: Schwann cells, satellite cells

Glial Cell Functions

- **Schwann cells** (施万细胞): Myelinate peripheral axons
- **Oligodendrocytes** (少突胶质细胞): Myelinate CNS axons
- **Astrocytes** (星形胶质细胞): Multiple support functions
- **Microglia** (小胶质细胞): Phagocytic immune cells
- **Ependymal cells** (室管膜细胞): Line ventricles, produce CSF

Myelin Sheath Formation in PNS

- Schwann cell membrane wraps around axon multiple times
- Creates insulating myelin sheath
- **Neurilemma** (神经膜): Outer layer of Schwann cell
- Each Schwann cell myelinates one axon segment
- Critical for rapid nerve conduction

Myelinated vs Unmyelinated Axons

- Electron microscopy reveals structural differences
- Myelinated axons: Multiple membrane layers, Schwann cell cytoplasm outside
- Unmyelinated axons: Still associated with Schwann cells but no myelin wrapping
- Myelination dramatically increases conduction velocity

CNS Myelination

- **Oligodendrocytes** (少突胶质细胞) form myelin in CNS
- One oligodendrocyte myelinates multiple axons
- Multiple tentacle-like processes extend to different axons
- **White matter** (白质): Myelinated axons (appears white)
- **Gray matter** (灰质): Cell bodies and unmyelinated processes

Peripheral Nerve Regeneration

- PNS axons can regenerate after injury
- Schwann cells form **regeneration tube** (再生管)
- Guides regrowing axon to proper destination
- (a) Axon severed; (b) distal portion degenerates; (c-d) regrowth through tube; (e) reconnection
- CNS regeneration much more limited

Astrocyte Functions

- Astrocytes have **end-feet** (终足) on capillaries and synapses
- Take up glucose from blood, convert to lactate for neuronal energy
- Take up neurotransmitter **glutamate** (谷氨酸), convert to **glutamine** (谷氨酰胺)
- Recycle glutamine to neurons for neurotransmitter synthesis
- Essential for metabolic support of neurons

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Membrane Potentials

- **Resting membrane potential** (静息膜电位): Approximately −70 mV
- **Depolarization** (去极化): Membrane becomes less negative
- **Hyperpolarization** (超极化): Membrane becomes more negative
- Measured using intracellular and extracellular electrodes
- Basis for all electrical signaling in neurons

Recording Membrane Potential Changes

- Oscilloscope or computer displays voltage difference over time
- Intracellular electrode inside neuron, extracellular electrode outside
- Resting potential stable at −70 mV
- Depolarization moves toward 0 mV
- Hyperpolarization moves more negative than rest

Voltage-Gated Ion Channels

- **Voltage-gated channels** (电压门控通道) open/close with voltage changes
- Channel closed at rest
- Opens when membrane depolarizes to threshold
- Allows specific ions (Na^+ or K^+) to diffuse through
- Essential for action potential generation

Ion Channel Dynamics

- Two types critical for action potentials:
- **Voltage-gated Na^+ channels** (电压门控钠通道): Open rapidly at threshold
- **Voltage-gated K^+ channels** (电压门控钾通道): Open more slowly
- Sequential opening creates positive feedback (Na^+) then negative feedback (K^+)

Action Potential Waveform

- Top: Action potential reaches +30 mV then repolarizes
- Bottom: Na^+ influx during depolarization (red)
- K^+ efflux during repolarization (blue)
- Precise timing of ion movements generates characteristic waveform
- All-or-none response once threshold reached

All-or-None Law

- **All-or-none law** (全或无定律): Action potential either occurs fully or not at all
- Subthreshold stimuli: No action potential
- Threshold stimulus: Full action potential
- Suprathreshold stimuli: Same amplitude action potential
- Amplitude does not encode stimulus strength

Encoding Stimulus Intensity

- Stimulus intensity encoded by **frequency** (频率) not amplitude
- Weak stimuli: Low frequency or no response
- Moderate stimuli: Moderate frequency
- Strong stimuli: High frequency of action potentials
- Frequency coding allows graded information transmission

Refractory Periods

- **Absolute refractory period** (绝对不应期): No response possible
- Occurs during Na^+ channel inactivation
- **Relative refractory period** (相对不应期): Only strong stimuli effective
- Occurs during hyperpolarization phase
- Limits maximum firing frequency, ensures unidirectional propagation

Cable Properties

- **Cable properties** (电缆特性) allow local current spread
- Positive charge injection creates local depolarization
- Depolarization spreads decrementally to adjacent regions
- Decreases with distance due to membrane resistance and capacitance
- Allows action potential to trigger adjacent regions

Unmyelinated Axon Conduction

- Action potentials regenerate continuously along entire axon
- Each segment depolarizes adjacent segment
- Previously active region is refractory, ensuring forward propagation
- Relatively slow conduction (0.5-2 m/s)
- Found in small diameter fibers (pain, autonomic)

Saltatory Conduction

- **Saltatory conduction** (跳跃式传导) in myelinated axons
- Myelin insulates axon, prevents ion current
- Action potentials only at **nodes of Ranvier** (郎飞结)
- Current “jumps” from node to node
- Much faster conduction (up to 120 m/s)

Conduction Velocity

- Conduction velocity depends on:
- Axon diameter: Larger = faster (less resistance)
- Myelination: Myelinated = much faster
- Type A fibers: Large, myelinated, fast (motor, proprioception)
- Type C fibers: Small, unmyelinated, slow (pain, autonomic)

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Types of Synapses

- **Electrical synapses** (电突触): Direct coupling via gap junctions
- **Chemical synapses** (化学突触): Communication via neurotransmitters
- Electrical: Bidirectional, very fast, found in cardiac and smooth muscle
- Chemical: Unidirectional, slower, allows for modulation and integration

Gap Junctions

- **Gap junctions** (缝隙连接) form electrical synapses
- Composed of **connexin** (连接蛋白) proteins forming **connexons** (连接子)
- Create water-filled channels between cells
- Ions pass directly from cell to cell
- Allows rapid, synchronized activity

Chemical Synapse Structure

- **Presynaptic neuron** (突触前神经元): Releases neurotransmitter
- **Synaptic cleft** (突触间隙): 20-50 nm gap
- **Postsynaptic neuron** (突触后神经元): Responds to neurotransmitter
- **Synaptic vesicles** (突触小泡): Store neurotransmitter
- Visible by electron microscopy

Neurotransmitter Release

- Action potential opens voltage-gated Ca^{2+} channels
- Ca^{2+} enters terminal, binds **synaptotagmin** (突触结合蛋白)
- **SNARE proteins** (SNARE 蛋白) mediate vesicle docking
- Vesicle fuses with membrane (**exocytosis**, 胞吐)
- Neurotransmitter released into synaptic cleft

Functional Specialization

- Dendrites and cell body: Integration region
- Contains **ligand-gated channels** (配体门控通道)
- Receives EPSPs and IPSPs, integrates signals
- Axon hillock: Action potential initiation site
- Axon: Conduction region with **voltage-gated channels** (电压门控通道)

Excitatory Synaptic Transmission

- Neurotransmitter released from presynaptic terminal
- Binds to ligand-gated channels on dendrites/cell body
- Opens channels causing ion flow
- Produces graded **EPSP** (兴奋性突触后电位)
- If threshold reached at axon hillock, action potential generated

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Acetylcholine Overview

- **Acetylcholine (ACh)** (乙酰胆碱) major neurotransmitter in PNS and CNS
- Synthesized from **choline** (胆碱) and **acetyl-CoA** (乙酰辅酶 A)
- Enzyme: **Choline acetyltransferase** (胆碱乙酰转移酶)
- Two receptor types: Nicotinic and muscarinic
- Found at neuromuscular junctions, ganglia, and in brain

Nicotinic ACh Receptors

- **Nicotinic receptors** (烟碱型受体) are ligand-gated ion channels
- (a) Closed when no ACh bound
- (b) Opens when ACh binds, allowing Na^+ and K^+ flow
- Na^+ influx predominates, causing depolarization
- Produces **EPSP** (兴奋性突触后电位)

Muscarinic ACh Receptors

- **Muscarinic receptors** (毒蕈碱型受体) are G-protein coupled
- ACh binding causes G-protein dissociation
- Beta-gamma subunits activate K^+ channels
- K^+ efflux causes hyperpolarization
- Example: Slows heart rate in cardiac pacemaker cells

Acetylcholinesterase

- **Acetylcholinesterase (AChE)** (乙酰胆碱酯酶) terminates ACh signal
- Hydrolyzes ACh into acetate and choline
- Occurs in synaptic cleft
- Choline recycled back to presynaptic terminal
- Resynthesized into ACh for reuse

Graded Postsynaptic Potentials

- **EPSPs** (兴奋性突触后电位) are graded, not all-or-none
- Stimulus strength determines EPSP amplitude
- Weak stimuli: Small EPSPs below threshold
- Strong stimuli: Larger EPSPs reaching threshold
- Threshold crossing triggers action potentials

Action Potentials vs EPSPs

- Action potentials: All-or-none, propagated, in axons
- EPSPs: Graded, decremental, in dendrites/soma
- Action potentials: Have refractory period
- EPSPs: No refractory period, can summate
- Both essential for neural signaling

Drugs Affecting Neuromuscular Transmission

- **Botulinum toxin** (肉毒杆菌毒素): Blocks ACh release
- **Tetanus toxin** (破伤风毒素): Blocks inhibitory neurotransmitters
- **Curare** (箭毒): Blocks nicotinic receptors
- **Nerve gases** (神经毒气): Inhibit AChE
- **Saxitoxin** (石房蛤毒素): Blocks Na^+ channels

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Monoamine Neurotransmitters

- **Monoamines** (单胺类) derived from amino acids
- **Catecholamines** (儿茶酚胺): Dopamine, norepinephrine, epinephrine
- **Indolamines** (吲哚胺): Serotonin (5-HT)
- Involved in mood, motor control, arousal
- Targets for many psychiatric medications

Monoamine Metabolism

- Synthesized in presynaptic terminal
- Released by exocytosis
- Removed primarily by **reuptake** (再摄取)
- Degraded by **monoamine oxidase (MAO)** (单胺氧化酶)
- Cycle of synthesis, release, reuptake, and degradation

G-Protein Signaling

- Norepinephrine binds G-protein coupled receptor
- G-protein dissociates into alpha and beta-gamma subunits
- Alpha subunit activates **adenylate cyclase** (腺苷酸环化酶)
- Produces **cAMP** (环磷酸腺苷) as **second messenger** (第二信使)
- cAMP activates **protein kinase** (蛋白激酶)

G-Protein Activation Steps

- Step 1: Neurotransmitter binds receptor
- Step 2: G-protein activates, binds GTP
- Step 3: Alpha subunit dissociates, activates/inhibits enzyme
- Step 4: GTP hydrolyzed to GDP
- Step 5: G-protein reassociates

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GABA and Inhibition

- **GABA** (γ-氨基丁酸) is major inhibitory neurotransmitter in CNS
- GABA receptors contain Cl^- channel
- GABA binding opens channel
- Cl^- influx causes hyperpolarization
- Produces **IPSP** (抑制性突触后电位)

Neurotransmitter Diversity

- Acetylcholine
- Monoamines: Dopamine, norepinephrine, serotonin
- Amino acids: Glutamate (excitatory), GABA, glycine (inhibitory)
- Polypeptides: Substance P, endorphins, neuropeptide Y
- Endocannabinoids, gases (NO, CO), purines (ATP, adenosine)

Outline

Spatial Summation

- **Spatial summation** (空间总和): Multiple synapses active simultaneously
- One presynaptic neuron: Small EPSP, below threshold
- Two presynaptic neurons: EPSPs summate
- Combined depolarization reaches threshold
- Triggers action potential in postsynaptic neuron

Postsynaptic Inhibition

- **IPSP** (抑制性突触后电位) hyperpolarizes membrane
- Makes inside more negative than resting potential
- Increases distance to threshold
- Harder for EPSPs to trigger action potential
- Balance of EPSPs and IPSPs determines neuronal output

Integration and Plasticity

- **Synaptic integration** (突触整合): Combining multiple inputs
- **Convergence** (会聚): Multiple inputs to one neuron
- **Divergence** (发散): One neuron to multiple targets
- **Temporal summation** (时间总和): Repeated stimulation summates over time
- **Synaptic plasticity** (突触可塑性): Long-term changes in synaptic strength
- Basis for learning and memory

Integration and Plasticity

End of Chapter 7

Thank you for your attention!